SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**
Question Count: 5
Duration :60 Minutes

Code of Conduct

I D.Raghava 192525304 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.Raghava

Assessment Tasks
Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data.
Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. **Agriculture Smart Farming Problem**
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem

A retail company has data coming from physical stores, its online shopping platform, inventory systems, and customer activities. Since the amount of data is very large, a big data architecture is required to process the information efficiently and provide useful insights.

Proposed Big Data Architecture

The system can be designed using the following layers:

Data Sources → Data Ingestion → Data Storage → Data Processing → Analytics & Machine Learning → Applications

1. Data Sources

Data can be collected from:

- Point-of-sale systems in physical stores
- Online shopping websites and mobile applications
- Inventory and warehouse systems
- Customer browsing and purchase history
- Loyalty programs
- Product reviews and feedback

2. Data Ingestion

Tools such as **Apache Kafka** can be used to collect real-time sales and inventory events. For batch data, tools such as **Apache Sqoop** or cloud-based data ingestion services can be used.

Kafka is useful because it can handle a large number of events continuously without stopping the entire system.

3. Data Storage

The company can store large amounts of structured and unstructured data in a data lake using technologies such as **Hadoop HDFS, Amazon S3, Azure Data Lake, or Google Cloud Storage.**

A data warehouse such as **Snowflake, BigQuery, or Amazon Redshift** can be used for structured business reports and dashboards.

4. Data Processing

Apache Spark can process both historical and real-time data. Spark can analyze sales trends, customer behavior, stock levels, and product performance.

For real-time inventory tracking, Spark Streaming or another stream-processing system can process incoming sales transactions immediately.

5. Demand Forecasting

Machine learning models can be trained using historical sales, seasonal trends, holidays, promotions, and customer demand.

Suitable models include:

- Linear Regression for simple forecasting
- Random Forest for demand prediction
- XGBoost for more accurate prediction
- LSTM models for time-series forecasting

For example, if sales of umbrellas increase before and during rainy periods, the system can predict higher future demand and recommend increasing stock.

6. Personalized Marketing

Customer purchase history and browsing behavior can be analyzed using recommendation algorithms.

Collaborative filtering can recommend products based on similar customers.

Content-based filtering can recommend products similar to items a customer has previously purchased or viewed.

7. Expected Benefits

The proposed architecture helps the company:

- Track inventory in real time
- Reduce stock shortages and excess inventory
- Forecast future demand
- Provide personalized product recommendations
- Improve customer satisfaction
- Increase sales and reduce operational costs

Simple Architecture

Stores / Website / Mobile App
↓
Apache Kafka – Data Ingestion
↓
Data Lake – HDFS / S3
↓
Apache Spark – Processing

↓
Machine Learning – Forecasting & Recommendation
↓
Dashboards / Inventory System / Personalized Offers

2. Social Media Fraud Detection Problem

A social media platform generates billions of posts, comments, likes, shares, and other interactions every day. Processing all this information requires a highly scalable system. The platform also needs to detect fake accounts and misinformation as quickly as possible.

Proposed Big Data Solution

The architecture can be divided into the following stages:

User Activity → Stream Processing → Data Storage → Feature Extraction → Machine Learning → Alerts/Actions

1. Data Collection

Data can be collected from:

- Posts
- Comments
- Likes
- Shares
- Login information
- Account creation details
- IP and device information
- User interaction patterns

2. Stream Processing

Apache Kafka can receive billions of user events continuously.

Apache Flink or Spark Streaming can process these events in real time.

For example, if an account suddenly sends thousands of similar messages within a short period, the streaming system can immediately identify the unusual behavior.

3. Data Storage

Historical information can be stored in a distributed data lake using **HDFS, Amazon S3, or another cloud storage platform.**

Databases such as **Cassandra** can also be used when very large-scale and fast access to user-related data is required.

4. Machine Learning for Fake Account Detection

Machine learning can identify patterns that are difficult to detect manually.

Useful features include:

- Number of followers and following
- Account age
- Posting frequency
- Number of interactions
- Similarity between posts
- Login locations
- Device usage
- Unusual activity patterns

Models such as **Random Forest, XGBoost, Logistic Regression, and neural networks** can be used to classify accounts as genuine or suspicious.

5. Misinformation Detection

Natural Language Processing (NLP) can analyze the content of posts and identify potentially misleading information.

Models such as **BERT or other transformer-based NLP models** can analyze text and classify content based on previously trained examples.

The system can also use:

- Text similarity
- Source credibility
- User reporting
- Image analysis
- Network or propagation patterns

6. Real-Time Detection

A real-time pipeline can work as follows:

User Post
↓
Kafka
↓
Flink / Spark Streaming
↓
Feature Extraction
↓
ML Model
↓
Risk Score
↓
Alert / Review / Restriction

For example, a suspicious account can receive a high fraud score. The platform can then send the account for additional verification or human review.

7. Benefits

This solution provides:

- Fast fraud detection
- Scalable processing
- Reduced fake-account activity
- Faster identification of misinformation
- Continuous monitoring
- Better platform security

Machine learning provides the intelligence, while stream-processing frameworks provide the speed needed for real-time detection.

3. Government Data Platform Problem

A government data platform needs to combine information from census, economic, geographic, and other departmental systems. Since many departments use different technologies and formats, proper data governance and interoperability are essential.

Proposed Data Management Strategy

The platform should follow the principle:

Data Sources → Data Integration → Central Data Platform → Governance & Security → Authorized Data Access

1. Data Integration

Different departments may store information in different formats such as CSV, JSON, XML, databases, and GIS formats.

Tools such as **Apache NiFi, Apache Kafka, and ETL/ELT** platforms can be used to collect and integrate this data.

Standard APIs should also be provided so that different government departments can exchange information.

2. Central Data Platform

A government data lake can store large datasets using technologies such as **Hadoop HDFS or cloud object storage.**

A data warehouse can be used for structured reporting and analysis.

Geospatial information can be managed using databases such as **PostgreSQL with PostGIS.**

3. Data Standards and Interoperability

All departments should follow common standards for:

- Data formats
- Names and definitions
- Geographic information
- Metadata
- APIs
- Data exchange

A common data dictionary should be maintained so that different departments understand the meaning of each data field in the same way.

4. Data Governance

A dedicated data governance framework should define:

- Who owns each dataset
- Who can access it
- How data can be used
- How long data should be retained
- How data quality is maintained
- How changes are recorded

A **data catalog** can help departments discover available datasets and understand their source and quality.

5. Privacy and Security

Government data may contain sensitive citizen information. Therefore, strong security controls are required.

The platform should use:

- Encryption during storage and transmission
- Role-Based Access Control (RBAC)
- Multi-factor authentication
- Data masking
- Anonymization or pseudonymization
- Audit logs
- Regular security monitoring

Only authorized users should have access to sensitive information.

6. Data Quality

Automated validation should identify:

- Missing values
- Duplicate records
- Incorrect formats
- Inconsistent information
- Outdated records

Data quality dashboards can help administrators monitor the overall quality of government datasets.

7. Expected Benefits

The proposed strategy provides:

- Better data sharing between departments
- Improved interoperability
- More reliable government data
- Strong privacy protection
- Better decision-making

- Reduced duplication of data
- Easier development of citizen services

4. Banking Fraud Detection Problem

Banks process millions of transactions every minute. Fraud detection must therefore happen with very low delay while maintaining high accuracy. The system should combine historical transaction information with real-time transaction streams.

Proposed Big Data Pipeline

Transaction → Kafka → Stream Processing → Feature Generation → Fraud Detection Model → Risk Score → Decision

1. Transaction Data

The system collects information such as:

- Account number
- Transaction amount
- Time
- Location
- Merchant
- Device
- Transaction type
- Previous transaction history

Historical transactions are stored in a distributed data lake or warehouse.

2. Real-Time Data Ingestion

Apache Kafka can receive transactions continuously.

Kafka is suitable because it can handle a very high number of events and distribute them across multiple processing systems.

3. Stream Processing

Apache Flink or Spark Streaming can analyze transactions as they arrive.

For example, suppose a customer normally spends ₹2,000–₹5,000 in one location. Suddenly, a transaction of ₹2,00,000 occurs from another location. The system can immediately identify this as unusual.

4. Feature Engineering

Important features include:

- Transaction amount
- Transaction frequency
- Time between transactions
- Customer spending pattern
- Geographic distance
- Device information
- Merchant category
- Previous fraud history

These features are provided to the machine learning model.

5. Fraud Detection Algorithms

Several algorithms can be combined.

Random Forest or XGBoost can classify transactions as legitimate or fraudulent**.**

Logistic Regression can provide a simpler and interpretable baseline model.

Isolation Forest can detect unusual transactions that do not look similar to normal transactions.

Deep learning models can also be used when a large amount of historical transaction data is available.

6. Real-Time Decision

The model generates a fraud probability or risk score.

For example:

Low Risk → Approve Transaction

Medium Risk → Request Additional Authentication

High Risk → Block/Review Transaction

7. Model Improvement

The system should continuously learn from confirmed fraud cases. New transaction information can be used to retrain the model so that fraud detection improves over time.

8. Security and Reliability

The system should use:

- Encryption

- Access control
- Audit logs
- Secure APIs
- Fault-tolerant processing
- Replication
- Backup systems

Simple Pipeline

Bank Transactions
↓
Apache Kafka
↓
Apache Flink / Spark Streaming
↓
Feature Engineering
↓
ML Fraud Detection Model
↓
Risk Score
↓
Approve / Verify / Block

This architecture provides both high processing speed and accurate fraud detection.

5. Agriculture Smart Farming Problem

Modern farms can generate large amounts of data from soil sensors, weather stations, drones, satellites, and farm equipment. Big data technology can combine this information and provide farmers with useful recommendations.

Proposed Precision Agriculture Architecture

Sensors / Drones / Weather Data → IoT Gateway → Data Ingestion → Data Storage → Analytics & ML → Farmer Dashboard

1. Data Collection

The system can collect:

- Soil moisture
- Soil temperature
- Soil nutrients

- Weather conditions
- Rainfall
- Crop health
- Pest information
- Drone images
- Satellite images
- Irrigation information

IoT sensors can continuously transmit information from different parts of a farm.

2. Data Ingestion

An IoT gateway can collect sensor information.

MQTT can be used for lightweight communication between sensors and the cloud.

Apache Kafka can be used when the system needs to handle a large number of continuous events.

3. Data Storage

Sensor and historical data can be stored in a cloud data lake.

Time-series databases can store frequently changing sensor readings.

Images captured by drones can be stored in object storage such as **Amazon S3, Azure Blob Storage, or Google Cloud Storage.**

4. Data Processing

Apache Spark can process large amounts of historical sensor and crop data.

Real-time processing can identify situations such as very low soil moisture and immediately notify the farmer.

5. Analytics

Different types of analytics can be applied.

Descriptive analytics can show current soil and weather conditions.

Predictive analytics can predict crop yield, rainfall, pest attacks, and irrigation requirements.

Prescriptive analytics can recommend what action the farmer should take.

6. Machine Learning

Machine learning can be used for:

- Crop yield prediction
- Disease detection
- Pest detection
- Irrigation prediction
- Fertilizer recommendation
- Crop selection

For example, an image classification model can analyze drone images and identify unhealthy areas of a crop.

Regression models can predict crop yield using soil, weather, and historical farming data.

7. Decision Support

The final information can be displayed through a mobile or web dashboard.

For example, the system may show:

Soil moisture is low → Irrigation recommended

High temperature + low humidity → Increase monitoring

Possible disease detected → Inspect affected area

This allows farmers to make decisions based on actual field conditions rather than relying only on manual observation.

8. Benefits

The proposed system can:

- Increase crop yield
- Reduce water consumption
- Reduce fertilizer usage
- Detect diseases early
- Reduce farming costs
- Improve resource utilization
- Support sustainable agriculture

Simple Architecture

Soil Sensors + Weather Stations + Drones
↓
IoT Gateway / MQTT
↓
Kafka / Data Ingestion
↓
Cloud Data Lake
↓
Spark Analytics + Machine Learning
↓
Prediction & Recommendations
↓
Farmer Mobile/Web Dashboard

Conclusion

Big data technologies can solve different industry problems by combining large-scale data collection, distributed storage, real-time processing, and machine learning. Technologies such as Kafka, Spark, Flink, Hadoop, cloud storage, and machine learning models can help organizations make faster and more accurate decisions. The most important part is not only collecting data but also converting it into useful information while maintaining scalability, security, privacy, and reliability.